

INSTRUCTOR GUIDE

monil1_ML-Note

LECTURE OVERVIEW

Main objectives:

- Introduce fundamental concepts of Machine Learning.
- Distinguish between different types of ML approaches.
- Outline a basic Machine Learning project workflow.

Key learning outcomes:

- Learners will define Machine Learning and its core purpose.
- Learners will identify scenarios suitable for supervised, unsupervised, or reinforcement learning.
- Learners will articulate the main stages of an ML project.
- Learners will recognize common applications of ML in daily life.

TEACHING NOTES

Topic: Introduction to Machine Learning

Core concept: Machine Learning allows systems to learn from data and improve performance on tasks without explicit programming.

Key teaching points:

- What is ML: Algorithms that learn from data to make predictions or decisions.
- Types of ML: Supervised (learns from labeled data, e.g., classification, regression), Unsupervised (finds patterns in unlabeled data, e.g., clustering, association).
- Basic ML workflow: Data acquisition, data preprocessing, model selection, training, evaluation, prediction/deployment.

1 real-world example: Predictive text on a smartphone keyboard learns from your typing history to suggest the next word.

1 discussion question: Considering the types of ML, which approach do you think powers personalized recommendations on Netflix?

TEACHING STRATEGIES

Timing breakdown:

- 10 min: Introduction and What is ML.
- 20 min: Types of ML with examples.
- 15 min: Basic ML workflow.
- 5 min: Q&A and wrap-up.

Interactive activities:

- Quick Poll: Ask students to raise hands for which ML type they think applies to an example provided (e.g., image recognition).

Think-Pair-Share: Give a problem statement (e.g., detecting fraudulent transactions) and have students pair up to discuss.

Key teaching tips:

Use visual aids for workflow diagrams and ML type distinctions.

Emphasize intuition over complex mathematics for initial concepts.

Connect every concept back to a tangible real-world application.

ASSESSMENT

Quick check questions:

What is the primary difference between supervised and unsupervised learning?

Name one common task for supervised learning and one for unsupervised learning.

List two critical steps in a Machine Learning workflow before model training.

Discussion prompts:

Discuss the ethical implications of using ML for hiring decisions.

How might the availability or quality of data impact the success of an ML project?

Key exam topics:

Definitions of Machine Learning, AI, Data Science (high-level differentiation).

Characteristics and examples of Supervised, Unsupervised, and Reinforcement Learning.

Stages of an ML project lifecycle.

Real-world applications of ML.

